A Short Introduction to Taylor Series

GPT and Hein

October 26, 2026

Taylor polynomials replace a function by a polynomial near a chosen point. Polynomials are easy to evaluate, differentiate, integrate, and graph. Taylor's theorem also tells us how large the approximation error can be.

1 Taylor polynomials

Suppose that f has enough derivatives near $x = a$. The degree-n Taylor polynomial of f centered at a is

$$T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n. \tag{1}$$

It has the same value, slope, and first n derivatives as f at $x = a$. When $a = 0$, it is called a *Maclaurin polynomial*.

Theorem 1.1 (Taylor's theorem with an error term)**.** *If f has $n+1$ continuous derivatives between a and x, then*

$$f(x) = T_n(x) + R_n(x), \qquad R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \tag{2}$$

for some number c between a and x. In particular, if $|f^{(n+1)}(t)| \le M$ between a and x, then

$$|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}. \tag{3}$$

The remainder $R_n(x)$ is the difference between the function and its polynomial approximation. Formula (3) is what turns an approximation into a controlled approximation.

2 Big O notation

Near $x = 0$, the statement

$$g(x) = \mathcal{O}(x^m)$$

means that there are constants $C > 0$ and $\delta > 0$ such that

$$|g(x)| \le C|x|^m \qquad \text{whenever } |x| < \delta.$$

For example,

$$\sin x = x + \mathcal{O}(x^3)$$

means that the error in replacing $\sin x$ by x is at most a constant times $|x|^3$ when x is close to 0. Big O describes the *size of the error*; it is not a number and should not be evaluated as though it were one.

3 Four important approximations

The following formulas are centered at 0:

$$\sin x = x + \mathcal{O}(x^3), \qquad |\sin x - x| \le \frac{|x|^3}{6}, \tag{S}$$

$$\cos x = 1 - \frac{x^2}{2} + \mathcal{O}(x^4), \qquad \left|\cos x - 1 + \frac{x^2}{2}\right| \le \frac{|x|^4}{24}, \tag{C}$$

$$e^x = 1 + x + \mathcal{O}(x^2), \qquad |e^x - (1+x)| \le \frac{e}{2}x^2 \quad (|x| \le 1), \tag{E}$$

$$\log(1+x) = x + \mathcal{O}(x^2), \qquad |\log(1+x) - x| \le 2x^2 \quad (|x| \le \tfrac{1}{2}). \tag{L}$$

The constants in the last two bounds are convenient rather than best possible. They follow directly from Taylor's theorem by bounding the second derivative. For cosine, the coefficient is $-1/2$, because $\cos''(0) = -1$ and the quadratic Taylor coefficient is $\cos''(0)/2! = -1/2$.

Example 3.1. Approximate $e^{0.05}$ using $e^x \approx 1 + x$. We obtain

$$e^{0.05} \approx 1.05.$$

Since $|0.05| \le 1$, equation (E) gives

$$|e^{0.05} - 1.05| \le \frac{e}{2}(0.05)^2 < 0.0034.$$

The actual error is much smaller, but the bound guarantees the stated accuracy. For comparison, to eight decimal places,

$$e^{0.05} = 1.051271\ldots$$

4 Approximating sine

The first three nonzero Taylor polynomials for sine are

$$\begin{aligned} T_1(x) &= x, \\ T_3(x) &= x - \frac{x^3}{3!}, \\ T_5(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!}. \end{aligned}$$

Each added term makes the approximation flatter or more curved in exactly the way needed to match more derivatives of sine at the origin. For the degree-5 polynomial

$$p(x) = x - \frac{x^3}{6} + \frac{x^5}{120},$$

the values at $x = 0.1$ are

$$\sin(0.1) = 0.099833416646828\ldots, \qquad p(0.1) = 0.099833416666667\ldots.$$

They differ by less than 2×10^{-11}.

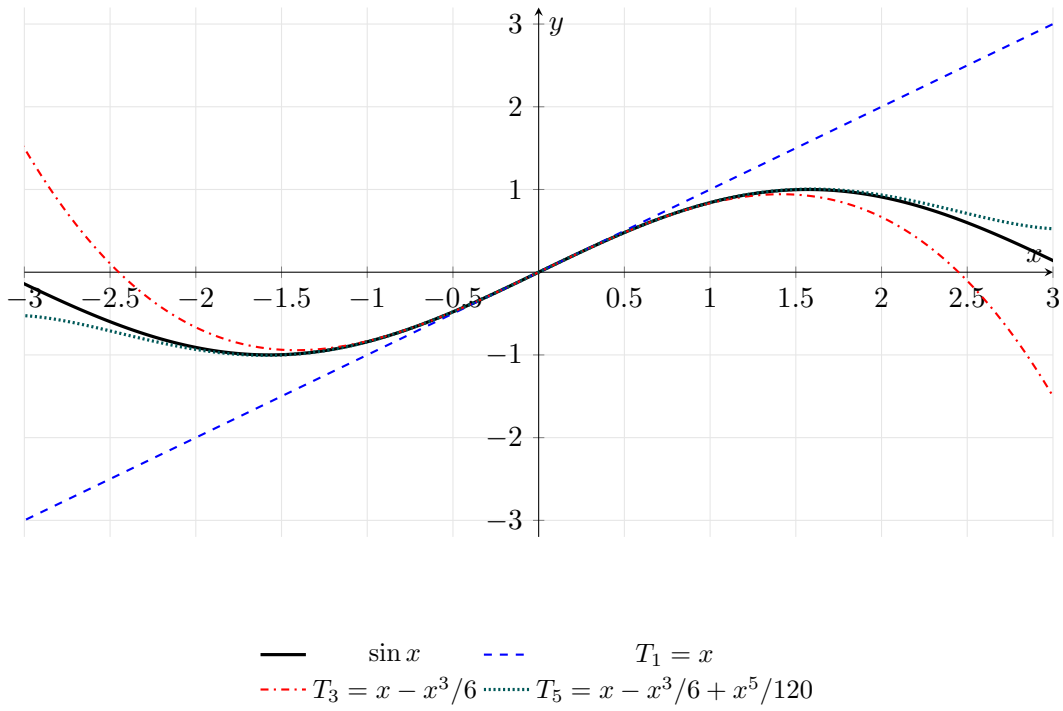


Figure 1: The sine function and its Taylor polynomials of degrees 1, 3, and 5. All four curves agree closely near $x = 0$.

The graph illustrates an important warning: a Taylor polynomial is usually a *local* approximation. Even a very good approximation near the center may be poor far away from it.

5 The complex exponential

Electrical engineers use $j = \sqrt{-1}$, so $j^2 = -1$. Substitute jx into the power series for the exponential and separate the even and odd powers:

$$\begin{aligned} e^{jx} &= 1 + jx + \frac{(jx)^2}{2!} + \frac{(jx)^3}{3!} + \cdots \\ &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + j\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) \\ &= \cos x + j \sin x. \end{aligned}$$

Therefore

$$\boxed{e^{jx} = \cos x + j \sin x}.$$

This identity connects exponential functions, trigonometry, and Fourier analysis.

6 Taylor approximation in two variables

Suppose that $f(x, y)$ has continuous second partial derivatives near (x_0, y_0). Its linear approximation, with a quadratic error term, is

$$\begin{aligned} f(x,y) &= f(x_0,y_0) + \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial f}{\partial y}(x_0,y_0)(y-y_0) \\ &\quad + \mathcal{O}\big((x-x_0)^2 + (y-y_0)^2\big)\,. \\ &= f(x_0,y_0) + \nabla(f)(x_0,y_0)\cdot((x,y)-(x_0,y_0)) \\ &\quad + \mathcal{O}\big((x-x_0)^2 + (y-y_0)^2\big)\,. \end{aligned}$$

Thus the tangent plane supplies the first approximation, while the error is at most a constant times the square of the distance from (x_0, y_0) when (x, y) is sufficiently close to that point.

7 Homework

Give a numerical error bound whenever an approximation is requested.

1. Find the degree-2 Maclaurin polynomial for e^x.

2. Use $\sin x = x + \mathcal{O}(x^3)$ to approximate $\sin(0.1)$. Use (S) to bound the error.

3. Use the quadratic Taylor polynomial for cosine to approximate $\cos(0.2)$. Bound the error using (C).

4. Use $\log(1+x) \approx x$ to approximate $\log(1.1)$ and use (L) to bound the error.

5. What does $f(x) = 3 + 2x + \mathcal{O}(x^2)$ tell you about $f(0)$ and $f'(0)$?

6. Find the degree-3 Maclaurin polynomial for $f(x) = 1/(1-x)$.

7. A function satisfies $f(2) = 5$, $f'(2) = -1$, and $f''(2) = 6$. Find its degree-2 Taylor polynomial centered at 2.

8. Use $e^x \approx 1 + x$ to approximate $e^{0.2}$. Use (E) to bound the error.

9. According to Figure 1, which polynomial is usually the best approximation to $\sin x$ near 0: T_1, T_3, or T_5? Why?

10. Find the degree-2 Taylor polynomial for $f(x) = \sqrt{x}$ centered at $a = 1$.

A Solutions to the homework

1. Since every derivative of e^x equals e^x,

$$T_2(x) = 1 + x + \frac{x^2}{2}.$$

2. $\sin(0.1) \approx 0.1$, and

$$|\sin(0.1) - 0.1| \leq \frac{0.1^3}{6} \approx 1.67 \times 10^{-4}.$$

3.

$$\cos(0.2) \approx 1 - \frac{0.2^2}{2} = 0.98, \qquad |R| \leq \frac{0.2^4}{24} \approx 6.67 \times 10^{-5}.$$

4. Since $1.1 = 1 + 0.1$, $\log(1.1) \approx 0.1$. The stated bound gives $|R| \leq 2(0.1)^2 = 0.02$.

5. Comparing with the first-degree Taylor polynomial gives $f(0) = 3$ and $f'(0) = 2$.

6. The first derivatives at zero are $1, 1, 2, 6$, so

$$T_3(x) = 1 + x + x^2 + x^3.$$

7.

$$T_2(x) = 5 - (x-2) + \frac{6}{2}(x-2)^2 = 5 - (x-2) + 3(x-2)^2.$$

8. $e^{0.2} \approx 1.2$, and

$$|e^{0.2} - 1.2| \leq \frac{e}{2}(0.2)^2 < 0.0544.$$

9. T_5 is usually best near 0 because it matches more derivatives of sine there. The graph shows it staying close to $\sin x$ on a wider interval than T_1 or T_3.

10. Here $f(1) = 1$, $f'(1) = 1/2$, and $f''(1) = -1/4$. Therefore

$$T_2(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2.$$